Project Report

Multi-Agent Gaming with the Board Game Diplomacy

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1 What is the problem you are solving?

Large language models (LLMs) are typically evaluated on complex question answering and logical reasoning. However, these benchmarks fail to capture multi-agent challenges like negotiation, collaboration, and strategic planning. We address this gap by using the board game `Diplomacy` to investigate LLM behavior in a competitive social environment.

By assigning each of the seven `Diplomacy` powers to a different LLM agent, we force them to build partnerships, communicate, and develop long-term tactics. This setup provides a robust evaluation environment for studying trust development, strategic decision-making, and time-based adapting.

Our main objective is to evaluate LLM performance in long-horizon multi-agent settings. Specifically, we investigate whether agents enhance their performance across games using context, long-term memory, and strategic adaptation to opponents' play styles. Thus, we explore memory and social reasoning as key determinants in negotiation-based environments.

2 What data will you use?

2.1 Where will you get it?

Our project uses self-generated simulation data rather than a traditional external dataset. Data originates from the `Diplomacy` engine during matches between seven LLM agents. We utilize structured game data (board states, supply centers), interaction data (negotiation messages, submitted orders), and agent-level context (diary summaries, memory records, and execution traces).

2.2 How will you gather it?

We collect data through controlled experiments varying model assignments and memory settings. Analyzing the resulting records yields metrics like move success, territorial growth, and alliance behavior, allowing us to assess how different configurations impact strategic performance over time.

3 How will you solve the problem?

3.1 What LLMs do you plan to use?

We build on a fork of the open-source GoodStartLabs/AI_Diplomacy repository, which wraps the `Diplomacy` game engine with an LLM agent system. The codebase's unified `BaseModelClient` interface allows each of the seven powers to be assigned a different model, enabling direct head-to-head benchmarking. The model selection balances cost and performance and may evolve with new releases; the current lineup is: Grok 4.1 Fast (xAI), Gemma 4 31B and Gemini 2.5 Flash Lite (Google), Qwen 3.5 27B and Qwen 3.6

Plus (Alibaba), GPT OSS 120B (OpenAI), and Claude Haiku 4.5 (Anthropic). MiniMax, Kimi K2.5, and MiMo models were excluded after poor performance in early testing.

3.2 Which methods do you plan to apply?

Structured agent state. Each power is governed by a `DiplomacyAgent` that maintains strategic goals, a relationship map over the other six powers (Enemy / Unfriendly / Neutral / Friendly / Ally), and a private diary of phase-prefixed reflections. All three are initialized and updated via LLM calls every phase.

Diary-based long-term memory (within a game). After every phase, each agent appends a structured diary entry summarizing its reasoning, the outcome of its orders, and any notable diplomatic events. At every Spring movement phase, an LLM-driven *consolidation pass* rewrites the diary, pruning redundant content while preserving strategically relevant history. This prevents unbounded context growth while still allowing agents to adapt based on past experience.

Cross-game memory. After a full game concludes, each agent's consolidated diary (capped to a fixed token limit to ensure scalability) and final relationship map are persisted and injected into the system prompt of the *next* game as prior experience. This allows an agent to carry strategic lessons, knowledge of opponents' playing styles, and calibrated trust scores across multiple sequential games, forming the self-improvement loop that distinguishes this project from single-game LLM evaluations.

To enable behavioral analysis, all private diary entries and outgoing messages are logged separately throughout every game. This creates a structured record that reveals divergences between an agent's internal reasoning and its external communication, the primary signal for detecting deceptive behavior. Each agent also maintains an explicit numeric trust score per opponent, updated every phase, which is persisted as part of the cross-game memory.

3.3 What is your idea for a multi-agent workflow?

Each game phase executes the following pipeline concurrently across all active powers:

1. **Negotiation** (movement phases): Powers exchange free-text messages in a round-robin fashion, choosing between private and global messages based on the board state and diary.

2. **Planning & diary update**: Each agent optionally generates a short strategic plan, then writes a diary entry capturing intentions and perceived threats.

3. **Order generation**: Each agent's context prompt (board state, history, goals, relationships, recent diary) is sent to the LLM; invalid orders are filtered before submission.

4. **Phase advance & memory update**: The engine resolves orders simultaneously. Agents write a post-phase diary entry; on Spring phases, a parallel consolidation pass condenses the full diary to keep context bounded.

This loop repeats until a power controls 18 supply centers, a year limit is reached, or powers draw.

4 How will you measure success?

The evaluation of this project will follow a dual-track approach, combining quantitative performance metrics with qualitative behavioral analysis to determine how effectively multi-agent LLM systems can navigate the complex social and strategic landscape of the game Diplomacy. Since our primary research interest lies in the agents' ability to self-optimize through competition and memory, the most critical metric will be the "Improvement Delta," the measurable shift in strategic efficacy and consistency between the initial game rounds and subsequent rounds where agents have access to their own reflected performance and updated memory.

Quantitatively, we will track standard game-theoretic indicators such as the number of supply centers controlled, the duration of survival per round, and final win rates across the scheduled matches. This allows us to observe how different models handle the massive action space of Diplomacy and whether their performance improves over time. To account for the limited sample size of planned matches, we will rotate starting powers across games to minimize positional bias. A key part of the quantitative evaluation will involve "Memory-Induced Success Rates," where we compare the performance of agents with persistent memory against a control baseline of "naive" agents (playing without memory loops) to isolate the impact of the self-reflection mechanism on win rates and alliance stability.

Qualitatively, the project will dive into behavioral characteristics, specifically focusing on the axis of "Honorable vs. Deceitful" gameplay. We will implement a system to analyze the delta between an agent's private reasoning (based on created artefacts like diaries and, where possible, internal chain-of-thought) and its external communication with other players. Success in this area will be measured by the agent's ability to recognize and adapt to betrayal. For instance, we will evaluate whether an agent correctly identifies a "deceitful" opponent based on past rounds and adjusts its trust parameters accordingly. This will be documented through "Trust Scores" that agents assign to one another in their memory logs, allowing us to visualize the evolution of diplomatic relations across the multi-round workflow.

A successful implementation will demonstrate a systematic shift in strategic behavior and an increase in nuanced, long-term planning as the agents progress through the 3 to 4 planned rounds (matches) of play. Success will be defined by our ability to produce a benchmark where the primary value comes from the agents' emerging ability to adapt, lie, and cooperate in ways that mimic human diplomatic complexity, specifically driven by the integration of their past experiences into their current decision-making.

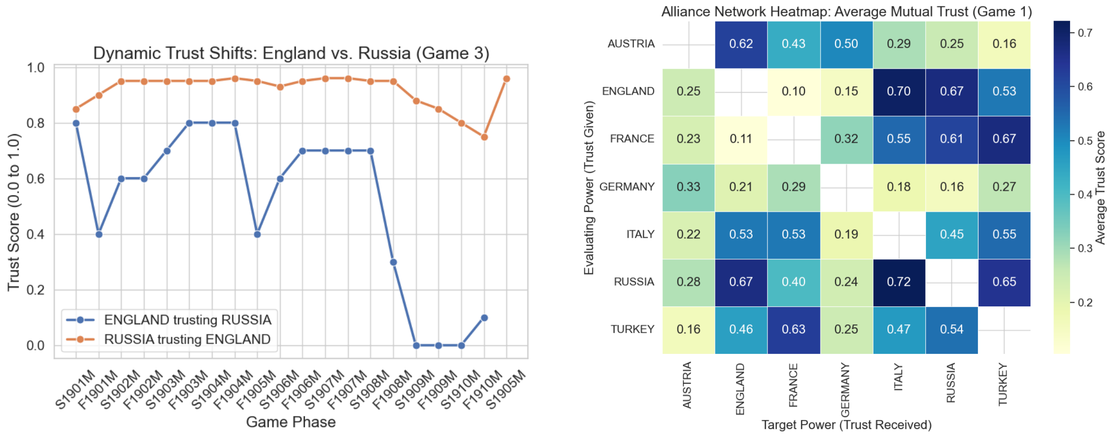

Figure 1: Core Capabilities of the LLM Trust System. The system captures localized, asymmetrical deception (left) while simultaneously generating coherent, board-wide diplomatic networks (right).

5 Results

5.1 Evaluation of the trust system

The following findings are derived from our first experiment, which evaluates the core trust system's behavior in "naive" agents. This establishes a baseline control group, allowing us to accurately isolate and measure the impact of the cross-game memory loops in subsequent experiments.

Dynamic Deception & The Sucker's Metric: Continuous tracking of mutual trust scores successfully quantified diplomatic backstabs. In Game 3, England and Russia maintained high mutual trust (0.7–0.9) until 1908, when England's internal trust in Russia plummeted to 0.0 ahead of a betrayal [1a]. Russia remained unaware, maintaining a 0.88 trust score. This visually proves the framework captures unilateral alliance breakdowns. Other extreme asymmetries included massive trust gaps of 0.95 (Game 1) and 0.90 (Game 2).

Trust vs. Game Success: Comparing average received trust against final supply centers revealed no meaningful correlation ($r = -0.05$), proving that universally trusted agents do not inherently win. Victory relies on tactical positioning and timely betrayals rather than simply projecting honesty. Additionally, we observed no strong "Paranoia Effect" ($r = 0.12$): agents that frequently perceived betrayal did not systematically withdraw trust across the board.

Global Alliance Mapping: By aggregating mutual trust scores into matrices, the system successfully mapped overarching geopolitical landscapes. Trust heatmaps from Game 1 clearly visualized strong mutual axes (England-Russia), isolated pariahs (Germany, Austria), and agents successfully playing multiple sides (Italy) [1b].

LLM Personas & Adaptation: Underlying architectures strictly dictate baseline diplomatic postures. Grok (`x-ai/grok-4.1-fast`) emerged as the most cooperative baseline (0.54), while Qwen variants were highly skeptical (∼0.33–0.42). However, these personas are not static. Progression tracking showed dynamic adaptation, with almost all models exhibiting a sharp, widespread decline in given trust during the late game (1908-1910) as board tensions peaked [2].

The Instruction-Following Artifact: Analysis of unprompted trust score updates revealed a massive architectural artifact rather than pure strategic paranoia. Models like Grok, Qwen, and GPT-OSS frequently failed to document their internal reasoning, making nearly 100% of their updates appear unprompted due to JSON schema adherence failures. Conversely, models like Claude and Gemini meticulously populated these textual evidence fields. Furthermore, this failure to document reasoning showed virtually no correlation ($r = -0.05$) with final game success.

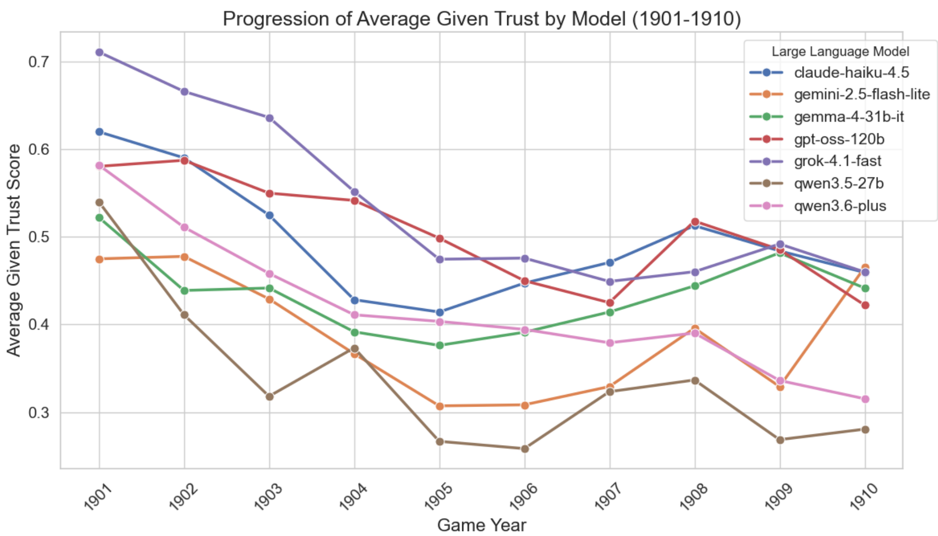

Figure 2: Average Given Trust by Model (1901-1910): Highlights baseline architectural differences and a sharp late-game decline across most models.

Ehrenwörtliche Erklärung

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Declaration of Used AI Tools

Tool	Purpose	Where?	Useful?
ChatGPT	Rephrasing	Throughout	+
DeepL	Translation	Throughout	+
ResearchGPT	Summarization of related work	Sec. **??**	-
Dall-E	Image generation	Figs. 2, 3	++
GPT-4	Code generation	functions.py	+
ChatGPT	Related work hallucination	Most of bibliography	++

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